

Nishant Singh

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SUMMARY

Lead Software Engineer with 7+ years of experience driving backend and platform architecture for high-growth products. Over the past 2.5+ years, set the technical direction for a 5-engineer backend and data platform team and drove cross-team architecture decisions across scheduling, notifications, pricing, and shared platform capabilities. Raised engineering standards through stronger design reviews, testing, and delivery discipline. Led one of the company's most important scaling efforts: Migrating off tableau to internal Agentic tooling built from ground up to support 1000+ users and move from deducing decisions from dashboards to actionable insights generated by LLM.

CORE SKILLS

Technical Leadership: Technical Direction, Design Reviews, Cross-Team Architecture, Multi-Team Delivery, Rollout Strategy, Engineering Standards, Product Partnership, Roadmap Ownership, Prioritization, Mentorship

Architecture & Systems: Distributed Systems, System Design, Backend Architecture, Platform Engineering, Service Decomposition, Microservices, Event-Driven Architecture, API Design, Scalability, Reliability, Performance Optimization, Caching

Backend & Platform: Python (FastAPI, Flask), SQL (PostgreSQL, MYSQL), NoSQL (DynamoDB), Redis, Kafka, SPARK

Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes, Terraform, Jenkins

AI & Data Systems: LLM Applications, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search, AI System Design, AWS Bedrock

PROFESSIONAL EXPERIENCE

Nike, CA

Dec 2023 - Present

Lead Software Engineer

Tech stack: *Python, Flask, FastAPI, Postgres, Redis, AWS, Docker, Kubernetes, llms*

Leading Nike BI Engineering team to support Consumer Analytics.

1. **Scalable Pipeline Architecture:** Built a YAML-driven Airflow config framework generating 100+ dynamic DAGs across Databricks and AWS, cutting pipeline onboarding to a single config change and saving **30% on Databricks compute costs**.
2. **Reusable ETL Frameworks:** Engineered a pluggable Python/Spark job library with 46 reusable classes for multi-cloud data ingestion and transformation, establishing a robust operational data framework and eliminating team-wide logic duplication.
3. **Data Quality Infrastructure:** Extended Nike's open-source *spark-expectations* DQ framework with automated two-sided aggregate range thresholds, error quarantining, and detailed run-statistics.
4. **High-Performance Production APIs:** Architected an asynchronous REST API (**FastAPI + asyncpg**) serving point-of-sale and inventory KPIs to 4,000+ users, featuring a 3-tier query cache with sub second latency that slashed database queries **by 80%**
5. **Access Control & Security:** Programmed automated role synchronization (AD and Databricks) and row-level security by geography and metric domain to manage granular user access matrixes. Ops cockpit **reduced** debugging from **hours to minutes**.
6. **Mentored** engineers through **design reviews**, pair programming sessions, and code quality assessments, establishing team-wide engineering standards and elevating execution quality

Amazon Web Services, Palo Alto, CA

Jun 2019 - Dec 2023

Data Engineer

Tech stack: *Spark, Python, Scala, AWS, EMR, LAMBDA, Type Script, Redshift*

- **Infrastructure Modernization:** Spearheaded migration from a legacy database to event-driven batch processing via **AWS Glue** and PySpark, reducing processing time for terabytes of financial data from **3 hours** to under **25 minutes**.
- **Productionizing Data Science Models:** Partnered with Data Scientists to refactor monolithic SageMaker machine learning notebooks into unit-testable PySpark workflows, cutting runtime from **6 hours** to under **2 hours**.
- **Cross-Functional Data Solutions:** Designed, developed, and optimized end-to-end data pipelines using SQL, Spark, and Redshift to convert complex sales workflow requirements into stable reporting environments.
- **Stakeholder Tools:** Built a Python/Flask/Redshift look-up tool used **100K+** times monthly to enable **cross-selling** and account management mapping across global business units.
- **Data Quality & Leadership:** Monitored, troubleshooted, and tuned complex distributed pipelines to resolve data quality anomalies and provided technical mentorship to junior engineers.

Automaton Anywhere, San Jose, CA

May 2018 - Aug 2018

Data Analyst Intern

Tech stack: *Spark, Python, AWS, docker*

- **DevOps Analytics & Optimization:** Analyzed deployment telemetry data to provide actionable recommendations that achieved 100% usability and increased efficiency across production CI/CD pipelines.
- **High-Throughput Data Processing:** Performance-tuned Jenkins data ingestion pipelines using Elastic Search indexing logic to process 5,000+ metrics per second, reducing data lag from hours to minutes.
- **Pipeline Reliability Engineering:** Analyzed Jenkins build logs to identify bottlenecks and eliminated execution failures, achieving 100% error-free pipeline execution across 50+ daily builds.
- **Real-Time Operations Dashboards:** Designed and deployed live dashboards for build performance monitoring, enabling data-driven decisions for strategic release planning and capacity management.
- **Cloud Automation:** Built automation bots for daily operational tasks across AWS and Google Cloud, increasing team productivity by 40% and eliminating manual toil.

EDUCATION

M.S. in Information Systems

Santa Clara University, CA

B.S. in Computer Science

Sikkim Manipal University, India